Python Lab Record (1-17)

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Outputs below were produced by actually running each program in a Linux workspace (not a Windows terminal).

CIRCLE AREA

```
airine@workspace:~/airine_python$ python circle_area.py
Enter radius:6.2
Area= 120.76282160399165
airine@workspace:~/airine_python$ _
```

LEAP YEAR

```
airine@workspace:~/airine_python$ python leapyear.py
Enter year: 2100
Not a leap year
airine@workspace:~/airine_python$ python leapyear.py
Enter year: 2028
Leap year
airine@workspace:~/airine_python$ _
```

LARGEST OF THREE

```
airine@workspace:~/airine_python$ python largestof3.py
Enter first number: 12
Enter second number: -7
Enter third number: 30
30 is the largest number
airine@workspace:~/airine_python$ _
```

FACTORIAL

```
airine@workspace:~/airine_python$ python factorial.py
Enter a number: 25
The factorial of 25 is 15511210043330985984000000
airine@workspace:~/airine_python$ _
```

FIBONACCI

```
airine@workspace:~/airine_python$ python fibonacci.py
Enter number of terms:9
0
1
1
2
3
5
8
13
21
airine@workspace:~/airine_python$ _
```

PRIME NUMBER

```
airine@workspace:~/airine_python$ python prime.py
Enter a number: 13
Prime number
airine@workspace:~/airine_python$ python prime.py
Enter a number: 15
Not a Prime number
airine@workspace:~/airine_python$ _
```

STRING

```
airine@workspace:~/airine_python$ python string.py
Enter a string:airine moyalan
Enter character to replace:a
Enter replacement character:o
The replaced string is: oirine moyolon
Reversed string is: nalayom eniria
Character frequency:
a : 3
i : 2
r : 1
n : 2
e : 1
  : 1
m : 1
o : 1
y : 1
l : 1
Number of words in the string: 2
airine@workspace:~/airine_python$ _
```

LIST

```
airine@workspace:~/airine_python$ python list.py
Enter numbers separated by space:9 14 3 22 6
Original list: [9, 14, 3, 22, 6]
Enter number to search:3
3 is present in the list
Enter number to update:14
Enter new number:40
Updated list: [9, 40, 3, 22, 6]
Even numbers in the list: [40, 22, 6]
Enter number to add to the list:17
List after adding [9, 40, 3, 22, 6, 17]
Enter number to remove from the list:6
List after removing [9, 40, 3, 22, 17]
Sorted list: [3, 9, 17, 22, 40]
Reversed list: [17, 22, 3, 40, 9]
airine@workspace:~/airine_python$ _
```

DICTIONARY

```
airine@workspace:~/airine_python$ python dict.py
Enter keys:p q r s
Enter values:10 25 7 31

Dictionary: {'p': 10, 'q': 25, 'r': 7, 's': 31}
Dictionary sorted by values: {'r': 7, 'p': 10, 'q': 25, 's': 31}
Enter keys to second dictionary:t u
Enter values to second dictionary:5 19
Merged dictionary: {'p': 10, 'q': 25, 'r': 7, 's': 31, 't': 5, 'u': 19}
Enter a string:moyalan
Character frequency: {'m': 1, 'o': 1, 'y': 1, 'a': 2, 'l': 1, 'n': 1}

Enter elements of first set1:2 4 6 8 10
Enter elements of second set2:4 5 6 7 8
Common elements in both sets: {8, 4, 6}
Unique elements: {2, 5, 7, 10}
airine@workspace:~/airine_python$ _
```

TUPLE

```
airine@workspace:~/airine_python$ python tuple.py
Enter numbers: 7,19,-4,33,8,8
List: ['7', '19', '-4', '33', '8', '8']
Tuple: ('7', '19', '-4', '33', '8', '8')
Total elements: 6
First element: 7
Last element: 8
Reversed list: ['8', '8', '33', '-4', '19', '7']
airine@workspace:~/airine_python$ _
```

STUDENT MARKS

```
airine@workspace:~/airine_python$ python student.py
Enter the number of students: 2

Enter details for Student 1
Enter student name: Airine
Enter number of subjects: 2
Enter mark for subject 1: 88
Enter mark for subject 2: 93

Enter details for Student 2
Enter student name: Meera
Enter number of subjects: 2
Enter mark for subject 1: 76
Enter mark for subject 2: 81

========== STUDENT PERFORMANCE ==========

Student Name : Airine
Marks        : [88.0, 93.0]
Total Marks  : 181.0
Average      : 90.5
Highest Mark : 93.0
Lowest Mark  : 88.0
Grade        : A+

Student Name : Meera
Marks        : [76.0, 81.0]
Total Marks  : 157.0
Average      : 78.5
Highest Mark : 81.0
Lowest Mark  : 76.0
Grade        : B

=========================================
airine@workspace:~/airine_python$ _
```

BANK ACCOUNT

```
airine@workspace:~/airine_python$ python bankacc.py
Enter account number: 1024
Enter account holder name: Airine Shaju Moyalan
Enter account type: Savings
Enter initial balance: 25000

--- Account Details ---
Account Number: 1024
Account Holder: Airine Shaju Moyalan
Account Type: Savings
Balance: 25000.0

Enter amount to deposit: 5000
Amount deposited: 5000.0
Enter amount to withdraw: 3200
Amount withdrawn: 3200.0

--- Account Details ---
Account Number: 1024
Account Holder: Airine Shaju Moyalan
Account Type: Savings
Balance: 26800.0
airine@workspace:~/airine_python$ _
```

WEATHER

```
airine@workspace:~/airine_python$ python Weather_JSON.py
City       Temp    Humidity
Kochi 30.8 80
Delhi 35.1 42
Mumbai 32.6 71
Bengaluru 26.9 58

Total Cities: 4
Average Temperature: 31.35
Average Humidity: 62.75
Hottest City: Delhi
Coldest City: Bengaluru
airine@workspace:~/airine_python$ _
```

ELECTRICITY BILL

```
airine@workspace:~/airine_python$ python bill.py
Enter units: 80
Total Bill: Rs. 400.0
Thank you!
airine@workspace:~/airine_python$ python bill.py
Enter units: 150
Total Bill: Rs. 850.0
Thank you!
airine@workspace:~/airine_python$ python bill.py
Enter units: 320
Total Bill: Rs. 2400.0
Thank you!
airine@workspace:~/airine_python$ _
```

IMAGE PROCESSOR

```
airine@workspace:~/airine_python$ python image.py
Original Image:
 [[ 15  55 105 155]
 [ 25  65 115 165]
 [ 35  75 125 175]
 [ 45  85 135 185]]
Reshaped:
 [[ 15  55 105 155  25  65 115 165]
 [ 35  75 125 175  45  85 135 185]]
Flattened:
 [ 15  55 105 155  25  65 115 165  35  75 125 175  45  85 135 185]
Centre Region:
 [[ 65 115]
 [ 75 125]]
First Row: [ 15  55 105 155]
Last Column: [155 165 175 185]
Bright Image:
 [[ 75 115 165 215]
 [ 85 125 175 225]
 [ 95 135 185 235]
 [105 145 195 245]]
Dark Image:
 [[  0   0  45  95]
 [  0   5  55 105]
 [  0  15  65 115]
 [  0  25  75 125]]
Horizontal Flip:
 [[155 105  55  15]
 [165 115  65  25]
 [175 125  75  35]
 [185 135  85  45]]
Vertical Flip:
 [[ 45  85 135 185]
 [ 35  75 125 175]
 [ 25  65 115 165]
 [ 15  55 105 155]]
airine@workspace:~/airine_python$ _
```

LIBRARY

```
airine@workspace:~/airine_python$ python library_Catalogue.py

1. Add Book
2. Search Book
3. Total Books
4. Books by Author
5. Exit
Enter choice: 1
Enter title: Wings of Fire
Enter author: APJ Abdul Kalam
Book added successfully

1. Add Book
2. Search Book
3. Total Books
4. Books by Author
5. Exit
Enter choice: 1
Enter title: Malgudi Days
Enter author: R K Narayan
Book added successfully

1. Add Book
2. Search Book
3. Total Books
4. Books by Author
5. Exit
Enter choice: 2
Enter title or author: narayan
Malgudi Days by R K Narayan

1. Add Book
2. Search Book
3. Total Books
4. Books by Author
5. Exit
Enter choice: 3
Total books: 2

1. Add Book
2. Search Book
3. Total Books
4. Books by Author
5. Exit
Enter choice: 4
Enter author name: apj
Wings of Fire by APJ Abdul Kalam

1. Add Book
2. Search Book
3. Total Books
4. Books by Author
5. Exit
Enter choice: 5
airine@workspace:~/airine_python$ _
```

INPUT VALIDATION

```
airine@workspace:~/airine_python$ python Input_validation.py
Enter email: airine@gmailcom
Invalid
Enter phone: 98765
Invalid
Enter log: TXAbcdefg
Transaction IDs: []
Total IDs: 0
airine@workspace:~/airine_python$ python Input_validation.py
Enter email: airine@example.com
Valid
Enter phone: 9876543210
Valid
Enter log: paid TXA12345678 and TXB87654321
Transaction IDs: ['TXA12345678', 'TXB87654321']
Total IDs: 2
airine@workspace:~/airine_python$ _
```